
A Comparative Study of EM Variants

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1 Introduction

1.1 Topic Selection

While reviewing the lecture notes for this course, I found the EM algorithm section particularly interesting. Notably, in the later slides, a few different variants of the EM algorithm were briefly mentioned, but not explored in depth. I decided to implement and compare these variants on a shared testbed to better understand their operational differences. So while there is some experimentation, I think it is closer to a reading project.

1.2 Problem Overview

The central challenge in latent variable models is that the marginal likelihood requires integrating out the unobserved variable U :

$$f_{\theta}(x) = \int f_{\theta}(x, u) du. \quad (1)$$

This integral appears inside the log in the maximum likelihood estimation (MLE) objective, making direct maximization infeasible. EM resolves this via successive approximation, but compute requirements can still be a bottleneck for more complex models.

2 Background

2.1 Maximum Likelihood Estimation with Latent Variables

Given observations X and latent variables U , the MLE problem is:

$$\hat{\theta}(x) \in \arg \max_{\theta \in \Theta} \log f_{\theta}(x), \quad f_{\theta}(x) = \int f_{\theta}(x, u) du. \quad (2)$$

The marginalization inside the log is the core difficulty.

2.2 The Evidence Lower Bound (ELBO)

For any auxiliary distribution $q(u)$, the log-likelihood decomposes as:

$$\log f_{\theta}(x) = \underbrace{\mathbb{E}_q[\log f_{\theta}(x, u)] - \mathbb{E}_q[\log q(u)]}_{F(q, \theta) \text{ (ELBO)}} + \text{KL}(q \parallel f_{\theta}(u \mid x)) \geq F(q, \theta). \quad (3)$$

Since $\text{KL} \geq 0$, the ELBO is always a lower bound on the log-likelihood. Setting $q = f_{\theta}(u \mid x)$ collapses the KL term to zero and makes the bound tight.

2.3 Computational Complexity

For a Gaussian Mixture Model with K components and n observations, the evidence requires summing over all K^n discrete assignment configurations:

$$m(x) = \sum_c p(c) \int f(\mu) \prod_{i=1}^n f(x_i | \mu_{c_i}) d\mu. \quad (4)$$

This scales exponentially in n , making exact inference strictly intractable.

3 Methods

All four methods alternate between an E-step (computing or approximating the posterior over latent states) and an M-step (updating θ). They differ in how the E-step and M-step are handled.

3.1 Base EM

Beginning with the base EM algorithm, we define:

$$\alpha_t(k) = f_\theta(x_1, \dots, x_t, Z_t = k), \quad (5)$$

$$\beta_t(k) = f_\theta(x_{t+1}, \dots, x_T | Z_t = k). \quad (6)$$

The posterior state marginals and pairwise marginals are

$$\gamma_t(k) = \frac{\alpha_t(k) \beta_t(k)}{\sum_j \alpha_t(j) \beta_t(j)}, \quad (7)$$

$$\xi_t(j, k) \propto \alpha_t(j) A_{jk} f_\theta(x_{t+1} | Z_{t+1} = k) \beta_{t+1}(k). \quad (8)$$

With $q^{(t)}(z) = f_{\theta^{(t-1)}}(z | x)$ the KL term in Eq. (3) vanishes, so the M-step maximizes $\mathbb{E}_{q^{(t)}}[\log f_\theta(x, U)]$, which separates over θ because all summations move outside the logarithm. The Lagrangian derivation extends to the chain-structured statistics γ_t, ξ_t , giving the closed-form updates

$$\hat{\pi}_k = \gamma_1(k), \quad \hat{A}_{jk} = \frac{\sum_t \xi_t(j, k)}{\sum_t \gamma_t(j)}, \quad \hat{\mu}_k = \frac{\sum_t \gamma_t(k) x_t}{\sum_t \gamma_t(k)}, \quad \hat{\sigma}_k^2 = \frac{\sum_t \gamma_t(k) (x_t - \hat{\mu}_k)^2}{\sum_t \gamma_t(k)}. \quad (9)$$

Algorithm 1 Base EM (Baum–Welch)

- 1: Initialize $\theta^{(0)}$
 - 2: **repeat**
 - 3: **E-step:** Run forward–backward to compute $\gamma_t(k), \xi_t(j, k)$ via Eqs. (7)–(8)
 - 4: **M-step:** Update $\theta^{(t)}$ via Eq. (9)
 - 5: **until** $|\log f_{\theta^{(t)}}(x) - \log f_{\theta^{(t-1)}}(x)| < \varepsilon$
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3.2 Variational EM (VEM)

VEM restricts q to a tractable family $\mathcal{Q}$ to handle settings where the exact posterior is intractable. Start by adopting the mean-field family $q(z) = \prod_{t=1}^T q_t(z_t)$, which discards the chain dependencies preserved by Eqs. (7)–(8). The E-step is the Coordinate Ascent Variational Inference (CAVI) sweep: cycling through $t = 1, \dots, T$, update

$$\log q_t(k) \propto \log \mathcal{N}(x_t; \mu_k, \sigma_k^2) + \sum_j q_{t-1}(j) \log A_{jk} + \sum_j q_{t+1}(j) \log A_{kj}, \quad (10)$$

which is the HMM specialization of the generic CAVI rule $\log q_j \propto \mathbb{E}_{-j}[\log f(\Theta_j, \Theta_{-j}, x)]$. After CAVI converges, the chain-aware γ_t, ξ_t in Eq. (9) are replaced by their mean-field surrogates

$$\tilde{\gamma}_t(k) = q_t(k), \quad \tilde{\xi}_t(j, k) = q_t(j) q_{t+1}(k), \quad (11)$$

and the M-step then applies Eq. (9) with these substitutions. Because q is restricted to $\mathcal{Q}$, the KL term in Eq. (3) does not vanish and VEM maximizes a strict lower bound on $\log f_\theta(x)$.

Algorithm 2 Variational EM (mean-field)

- 1: Initialize $\theta^{(0)}$ and $q^{(0)}(z) = \prod_t q_t^{(0)}(z_t)$
 - 2: **repeat**
 - 3: **E-step:** Sweep CAVI updates Eq. (10) until the ELBO converges; form $\tilde{\gamma}_t, \tilde{\xi}_t$ via Eq. (11)
 - 4: **M-step:** Update $\theta^{(t)}$ via Eq. (9) with $\tilde{\gamma}_t, \tilde{\xi}_t$ in place of γ_t, ξ_t
 - 5: **until** $|F(q^{(t)}, \theta^{(t)}) - F(q^{(t-1)}, \theta^{(t-1)})| < \varepsilon$
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3.3 Expectation conditional maximization (ECM)

ECM replaces the joint M-step of Eq. (9) with a sequence of conditional maximizations over parameter blocks. We partition θ into a dynamics block $\theta_d = (\pi, A)$ and a (μ, σ^2) block $\theta_e = (\mu_{1:K}, \sigma_{1:K}^2)$. The first CM-step uses γ_t, ξ_t from Eqs. (7)–(8) computed under $\theta^{(t-1)}$ and updates only the dynamics:

$$\hat{\pi}_k = \gamma_1(k), \quad \hat{A}_{jk} = \frac{\sum_t \xi_t(j, k)}{\sum_t \gamma_t(j)}. \quad (12)$$

We then refresh the E-step under $(\hat{\pi}, \hat{A}, \mu^{(t-1)}, \sigma^{(t-1)})$ to obtain new marginals γ'_t , and the second CM-step updates (μ, σ^2) using these refreshed posteriors:

$$\hat{\mu}_k = \frac{\sum_t \gamma'_t(k) x_t}{\sum_t \gamma'_t(k)}, \quad \hat{\sigma}_k^2 = \frac{\sum_t \gamma'_t(k) (x_t - \hat{\mu}_k)^2}{\sum_t \gamma'_t(k)}. \quad (13)$$

Each CM-step is a partial maximization of $F(q, \theta)$, so ECM inherits the monotone ascent of EM; the cost is one additional forward–backward pass per outer iteration.

Algorithm 3 Expectation Conditional Maximization (ECM)

- 1: Initialize $\theta^{(0)} = (\pi^{(0)}, A^{(0)}, \mu^{(0)}, \sigma^{(0)})$
 - 2: **repeat**
 - 3: **E-step 1:** Run forward–backward under $\theta^{(t-1)}$ to compute γ_t, ξ_t via Eqs. (7)–(8)
 - 4: **CM-step 1:** Update (π, A) via Eq. (12)
 - 5: **E-step 2:** Run forward–backward under $(\hat{\pi}, \hat{A}, \mu^{(t-1)}, \sigma^{(t-1)})$ to obtain γ'_t
 - 6: **CM-step 2:** Update (μ, σ^2) via Eq. (13)
 - 7: **until** $|\log f_{\theta^{(t)}}(x) - \log f_{\theta^{(t-1)}}(x)| < \varepsilon$
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3.4 Generalized EM (GEM)

GEM relaxes the M-step to require only a non-decreasing update of $F(q^{(t)}, \theta)$. We instantiate this with a damped step toward the Base EM maximizer $\hat{\theta}^{(t)}$ defined by Eq. (9). Here the update is

$$\eta^{(t)} = (1 - \alpha) \eta^{(t-1)} + \alpha \hat{\eta}^{(t)}, \quad \alpha \in (0, 1], \quad (14)$$

And any $\alpha \in (0, 1]$ yields a non-decreasing update and therefore a valid GEM step; $\alpha = 1$ recovers Base EM. We use $\alpha = 0.5$ throughout to expose the convergence-rate cost of an inexact M-step.

Algorithm 4 Generalized EM with damped M-step

Require: Damping coefficient $\alpha \in (0, 1]$

- 1: Initialize $\theta^{(0)}$
 - 2: **repeat**
 - 3: **E-step:** Run forward–backward to compute γ_t, ξ_t via Eqs. (7)–(8)
 - 4: **M-step:** Compute $\hat{\theta}^{(t)}$ via Eq. (9); update $\theta^{(t)}$ via Eq. (14)
 - 5: **until** $|\log f_{\theta^{(t)}}(x) - \log f_{\theta^{(t-1)}}(x)| < \varepsilon$
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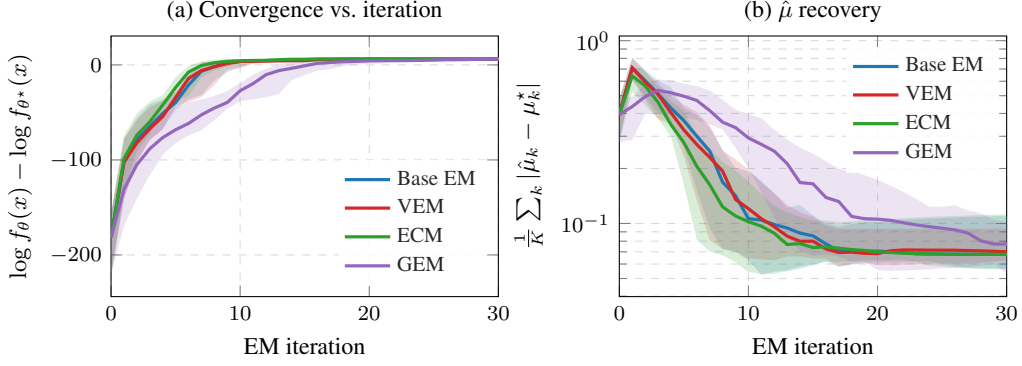


Figure 1: EM variants on Gaussian HMMs with random ground truth. (a) Excess log-likelihood. (b) $\hat{\mu}$ error.

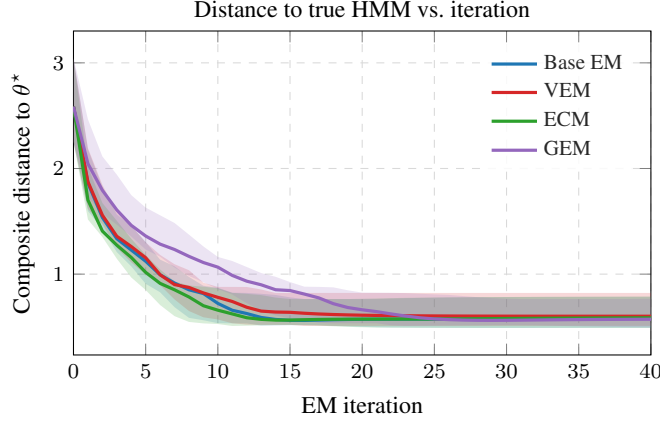


Figure 2: Distance to θ^* across iterations.

4 Experiments

Setup. We start by generating some HMMs, then simulate a large number of observations from each truth and run the four methods for from a shared initialization. Convergence is reported as the excess log-likelihood $\log f_\theta(x) - \log f_{\theta^*}(x)$ so that trajectories aggregate across the random truths.

Convergence and $\hat{\mu}$ recovery. Figure 1 shows convergence per iteration. Base EM, ECM, and VEM close the gap to the MLE within 10 iterations or so while GEM ($\alpha = 0.5$) takes roughly twice as long. The $\hat{\mu}$ error in panel (b) shows the log-likelihood trajectory for each method.

Distance to the true distribution. Figure 2 plots a composite distance between each recovered HMM and the ground truth.

Recovered marginal density. Figure 3 compares the stationary marginal $p_\theta(x) = \sum_k \pi_k^\infty \mathcal{N}(x; \mu_k, \sigma_k^2)$ under each method to the truth on a single random selection, with the empirical histogram of the observations as backdrop. Base EM, ECM, and VEM lay close to the ground truth, while GEM does not quickly close the distance.

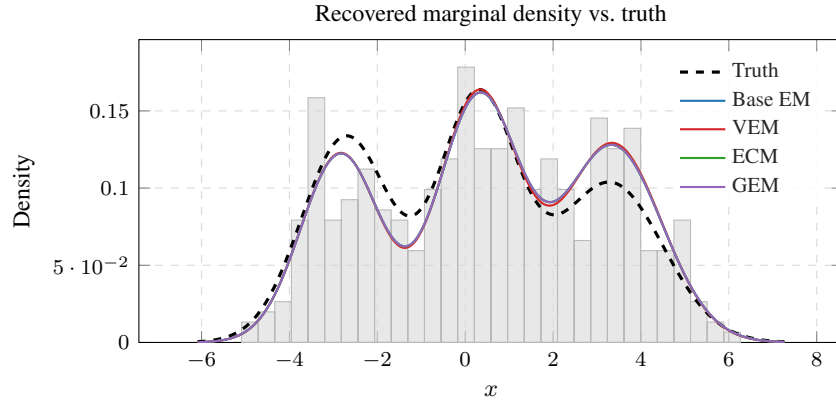


Figure 3: Recovered marginal density.

5 Discussion

Each of the four tested algorithms converges fairly quickly to the ground truth but not all at the same rate. Base EM, ECM, and VEM all reach the MLE within about 10 iterations, while GEM ($\alpha = 0.5$) takes roughly twice as long. I found that it is preferable to use Base EM when feasible, and to utilize a variant only algorithm when a harder model forces such a choice.